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Improving Pseudo-Multiview Learning via Dual-Sampler Bidirectional Feature Flow

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Abstract

Pseudo-multiview learning (PML) improves classification by generating complementary internal representations and fusing their evidence using subjective logic. However, conventional PML incurs substantial computational costs as the number of pseudo-views increases. Multiscale grid-based PML alleviates this issue by generating lower-resolution pseudo-views through downsampling and integrating cross-scale features. Nevertheless, its unidirectional top-down feature flow is vulnerable to noise in high-resolution branches, which can propagate to lower-resolution branches and degrade classification performance. This paper proposes a noise-resilient multiscale PML framework. The proposed dual-sampler module enables each branch to aggregate complementary features from both higher- and lower-resolution branches, thereby establishing bidirectional cross-scale information flow. In addition, a noise-aware spatial-channel attention module jointly models feature discriminativeness and reliability, suppressing corrupted spatial regions and feature channels during cross-scale fusion. Experimental results on real-world datasets from diverse domains, including BreakHis at 40×, 100×, 200×, and 400× magnifications, Oxford-IIIT Pet, and Chest X-ray, demonstrate the superiority of the proposed method over baseline models in classification accuracy, convergence stability, and robustness to cross-scale Gaussian noise.

Keywords: image classification; multiview learning; multiscale feature representation; bidirectional feature flow

MSC: 68T01

1. Introduction

Deep neural networks have achieved strong performance in computer vision tasks, including image classification, object detection, semantic segmentation, and medical image analysis. Recent models, such as ConvNeXt [1], Swin Transformer V2 [2], MaxViT [3], and ConvNeXt V2 [4], highlight the importance of hierarchical feature learning. Shallow layers capture local patterns, whereas deeper layers encode semantic information. However, useful visual cues often appear at different spatial scales. Fine cellular structures in histopathology images, localized abnormalities in chest X-rays, and subtle texture variations in natural images are representative examples. Multiscale learning is therefore widely used to improve visual recognition.

Multiscale architectures typically use feature pyramids or parallel branches with different resolutions. High-resolution features preserve fine details but require more computation and may be sensitive to local noise. Low-resolution features provide broader context at

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lower cost but may lose detailed information. Cross-scale feature fusion is therefore critical. Recent attention-based fusion methods improve the integration of complementary multiscale features [5,6]. However, a fusion strategy should also limit the transmission of irrelevant or corrupted information across branches.

Pseudo-multiview learning (PML) constructs multiple internal feature representations, termed pseudo-views, from a single input and combines their evidence using subjective logic [7]. This approach improves classification by exploiting complementary information and producing uncertainty-aware predictions. However, conventional PML requires separate feature extraction for each pseudo-view. Its computational and memory costs therefore increase with the number of pseudo-views.

Stability-enhanced PML addresses this issue using a multiscale grid architecture that generates lower-resolution pseudo-views through learnable downsampling [8]. It reduces the cost of independent pseudo-view generation and applies spatial-channel attention for cross-scale fusion. However, its feature flow is mainly top-down. Noise in high-resolution features can therefore propagate to lower-resolution branches and impair the final prediction. Moreover, it does not exploit low-resolution branches to support higher-resolution learning even though low-resolution features often converge faster.

This paper proposes an improved PML framework via dual-sampler bidirectional feature flow. The proposed dual-sampler module enables each branch to receive features from both higher and lower scales. This bidirectional flow allows stable low-resolution features to facilitate the learning of higher-resolution branches while retaining fine details from high-resolution features. In addition, this paper introduces a noise-aware spatial-channel attention module that considers feature reliability during fusion and suppresses corrupted information.

The main contributions are as follows:

- A dual-sampler module that enables bidirectional feature exchange between higher- and lower-resolution pseudo-view branches.
- A bidirectional flow that uses rapidly convergent low-resolution representations to support faster and more stable optimization of higher-resolution branches.
- A noise-aware spatial-channel attention module that suppresses unreliable features during cross-scale fusion.
- Experiments on the BreakHis, Oxford-IIIT Pet, and Chest X-ray datasets that show improved classification accuracy, convergence stability, calibration, and robustness to cross-scale Gaussian noise over the baseline methods.

2. Related Work

2.1. Multiview Learning

Multiview learning improves prediction by exploiting complementary information from multiple views of the same sample. In computer vision, these views may be acquired from different sensors, camera angles, feature spaces, or augmentations. Deep multiview methods usually learn view-specific representations and combine them through feature fusion, co-regularization, contrastive objectives, or late prediction fusion. Recent studies have further used transformers and attention mechanisms to model dependencies across views and to select informative view-specific features [9,10].

A related line of research uses multiscale feature representations in which feature maps at different spatial resolutions provide complementary information about the same input. Although these representations are not independent physical views, they can be interpreted as internal views of the input because each scale describes the sample from a different representational perspective. Several object-detection and segmentation architectures have demonstrated the value of bidirectional information exchange between such multiscale

representations. Path Aggregation Network (PANet) [11] extends the conventional top-down feature pyramid with an additional bottom-up path augmentation, improving the integration of fine-grained and high-level information. EfficientDet [12] further introduces a weighted bidirectional feature pyramid network (BiFPN), which repeatedly performs top-down and bottom-up cross-scale fusion while learning the relative importance of the incoming features.

The relationship between these approaches and multiview learning is conceptual rather than identical. PANet and BiFPN treat multiscale feature maps primarily as hierarchical representations for feature fusion, whereas multiview learning explicitly emphasizes the complementary evidence provided by multiple views. In particular, PANet and BiFPN do not generally construct separate class-level predictions for each scale or explicitly quantify the uncertainty and reliability of each representation. Nevertheless, their bidirectional cross-scale architectures provide an important basis for treating different resolutions as complementary internal views.

In practical applications, collecting multiple real views may be expensive or infeasible. Moreover, view quality may differ across samples. A corrupted or uninformative view can reduce the benefit of fusion if the model treats all views equally. These issues motivate the construction of reliable internal views from a single input and the use of adaptive fusion mechanisms.

2.2. Uncertainty-Aware Learning

Uncertainty-aware learning estimates not only class predictions but also the confidence or reliability of those predictions. This capability is important in applications involving noisy data, domain shifts, and safety-critical decisions, such as medical image analysis. Recent approaches estimate uncertainty through Bayesian approximations, deep ensembles, posterior modeling, or evidential learning [13–15].

Evidential learning predicts nonnegative evidence for each class and converts the evidence into the parameters of a Dirichlet distribution. The resulting distribution provides class probabilities and uncertainty without requiring repeated stochastic inference. Subjective logic further offers operators for combining evidence from multiple sources. This property makes subjective logic suitable for multiview learning, where each view may provide different levels of support for a prediction. However, evidence fusion remains sensitive to unreliable features or views. Therefore, feature reliability should be considered before evidence is aggregated.

Building upon this foundation, recent works leverage evidential uncertainty to address domain shifts and robust multi-source aggregation, such as through evidential active learning for sample selection under distribution shifts [16] or evidential weight aggregation for robust federated medical image segmentation [17].

2.3. Pseudo-Multiview Learning and Its Improved Version

Pseudo-multiview learning (PML) constructs multiple internal feature representations from a single input and treats them as pseudo-views [7]. Each pseudo-view produces class evidence, and subjective logic combines the evidence to generate an uncertainty-aware prediction. PML improves classification by capturing complementary information without requiring multiple physical sensors or viewpoints.

Despite this benefit, conventional PML uses separate branches to generate pseudo-views. Its computational cost and memory usage increase as more pseudo-views are added. Stability-enhanced PML addresses this limitation through multiscale grid feature extraction [8]. It produces lower-resolution pseudo-views by downsampling earlier features

and uses spatial-channel attention for cross-scale fusion. This design reduces redundant computation and improves feature interaction across scales.

However, stability-enhanced PML mainly propagates features from high-resolution branches to low-resolution branches. Noise in high-resolution features can thus spread through the grid and affect later pseudo-views. In addition, low-resolution branches are computationally compact and often converge faster, but their stable representations are not used to guide high-resolution branches.

2.4. Motivations

The limitations of the existing PML methods motivate a feature-flow strategy with two properties. First, feature exchange should be bidirectional. High-resolution branches provide detailed spatial cues, whereas low-resolution branches provide compact contextual representations and can converge earlier. Allowing information flow in both directions can improve feature learning across all the branches.

Second, cross-scale fusion should account for feature reliability. Standard spatial-channel attention emphasizes discriminative regions and channels, but it may still propagate noisy features if they receive high attention. A noise-aware attention mechanism can reduce this risk by suppressing unreliable spatial regions and channels before fusion.

Based on these observations, the proposed method introduces dual-sampler bidirectional feature flow and noise-aware spatial-channel attention. The method retains the efficiency of multiscale pseudo-view generation while improving convergence and robustness to cross-scale Gaussian noise.

3. Proposed Method

3.1. Overview

Figure 1 shows the proposed framework. Given an input image $\mathbf{x} \in \mathbb{R}^{3\times H\times W}$, an initial convolution and a pre-processing block (PPB) generate the first feature map. The network then constructs three pseudo-view branches at progressively lower resolutions. Each branch consists of residual dense blocks (RDBs), which extract discriminative features while preserving local and global information through dense residual connections.

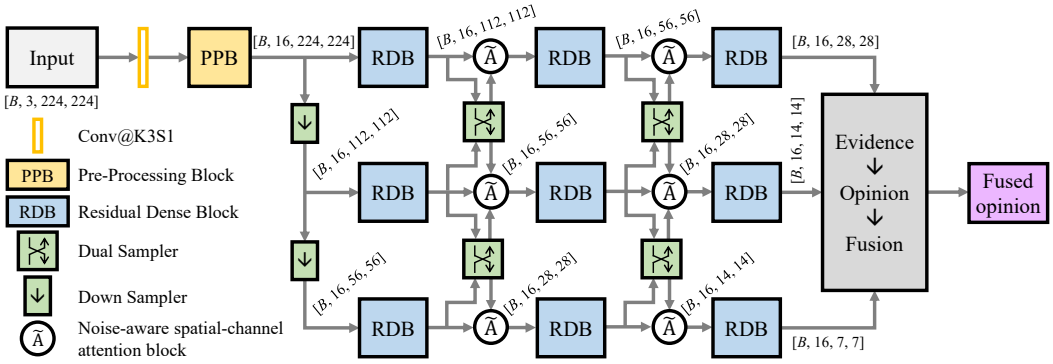


Figure 1. Overall architectural design of the proposed pseudo-multiview learning with dual-sampler bidirectional feature flow for three pseudo-views. Conv@K3S1 denotes a convolutional layer with kernel size 3×3 and stride 1. The notation $[B, C, H, W]$ denotes the batch size, channel number, height, and width.

The top, middle, and bottom branches operate at resolutions of 224×224, 112×112, and 56×56, respectively. Downsamplers generate the initial lower-resolution features. At each grid stage, the proposed dual-sampler module exchanges information between adjacent branches. The fused features are refined by the proposed noise-aware spatial-channel attention block (NA-SCAB) before entering the next RDB.

The output feature of the final RDB in each branch is transformed into a nonnegative evidence vector. Each evidence vector is represented as a subjective-logic opinion. The opinions from all the pseudo-views are then fused using the averaging rule to produce the final prediction.

3.2. Dual Sampling for Bidirectional Feature Flow

Let $\mathbf{F}_l^t \in \mathbb{R}^{C\times H_l\times W_l}$ denote the feature map of branch l at grid stage t, where $l \in \{1,2,3\}$ indicates the top, middle, and bottom branches, respectively. The spatial resolution decreases as l increases. A downsampler initializes the lower-resolution branches ,

$$\mathbf{F}_{l+1}^0 = \mathcal{D}\left(\mathbf{F}_l^0\right), \tag{1}$$

where $\mathcal{D}(\cdot)$ denotes a stride-2 convolution followed by feature transformation.

Existing multiscale PML propagates features only from higher to lower scales. Thus, noise in an upper branch may be transmitted to lower branches. In contrast, the proposed dual-sampler enables each branch to use information from both adjacent scales. As shown in Figure 2, it contains a downsampling path and an upsampling path. The downsampling path applies a stride-2 convolution to align a higher-resolution feature with the target scale. The upsampling path applies a transposed convolution to align a lower-resolution feature with the target scale.

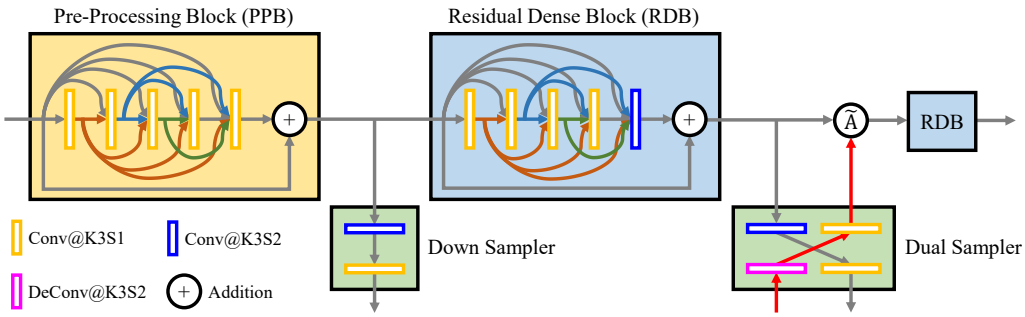


Figure 2. Architectural designs of pre-processing, residual dense, downsampler, and dual-sampler blocks. Connections are shown in different colors for ease of viewing.

For an intermediate branch, the dual-sampler output is defined as

$$\mathbf{Z}_l^t = \tilde{\mathcal{A}}(\mathbf{F}_l^t, \mathcal{D}(\mathbf{F}_{l-1}^t), \mathcal{U}(\mathbf{F}_{l+1}^t)), \quad l = 2, \tag{2}$$

where $\mathcal{U}(\cdot)$ denotes the upsampling operation and $\tilde{\mathcal{A}}(\cdot)$ denotes NA-SCAB. For the top and bottom branches, only the available adjacent feature is used:

$$\mathbf{Z}_1^t = \tilde{\mathcal{A}}(\mathbf{F}_1^t, \mathcal{U}(\mathbf{F}_2^t)), \tag{3}$$

$$\mathbf{Z}_3^t = \tilde{\mathcal{A}}(\mathbf{F}_3^t, \mathcal{D}(\mathbf{F}_2^t)). \tag{4}$$

The output feature at the next stage is obtained by

$$\mathbf{F}_l^{t+1} = \mathcal{R}_l^t(\mathbf{Z}_l^t), \tag{5}$$

where $\mathcal{R}_l^t(\cdot)$ denotes an RDB.

This design provides two benefits. First, high-resolution branches deliver fine spatial details to lower-resolution branches. Second, low-resolution branches provide compact contextual features to higher-resolution branches. Since low-resolution branches have smaller feature maps, they generally converge faster and can provide stable information during the

optimization of higher-resolution branches. Therefore, bidirectional flow improves both feature interaction and training stability.

3.3. Noise-Aware Spatial-Channel Attention

The proposed NA-SCAB suppresses unreliable features before they are propagated across branches. It consists of a noise-aware channel attention block (NA-CAB) and a noise-aware spatial attention block (NA-SAB), as shown in Figure 3.

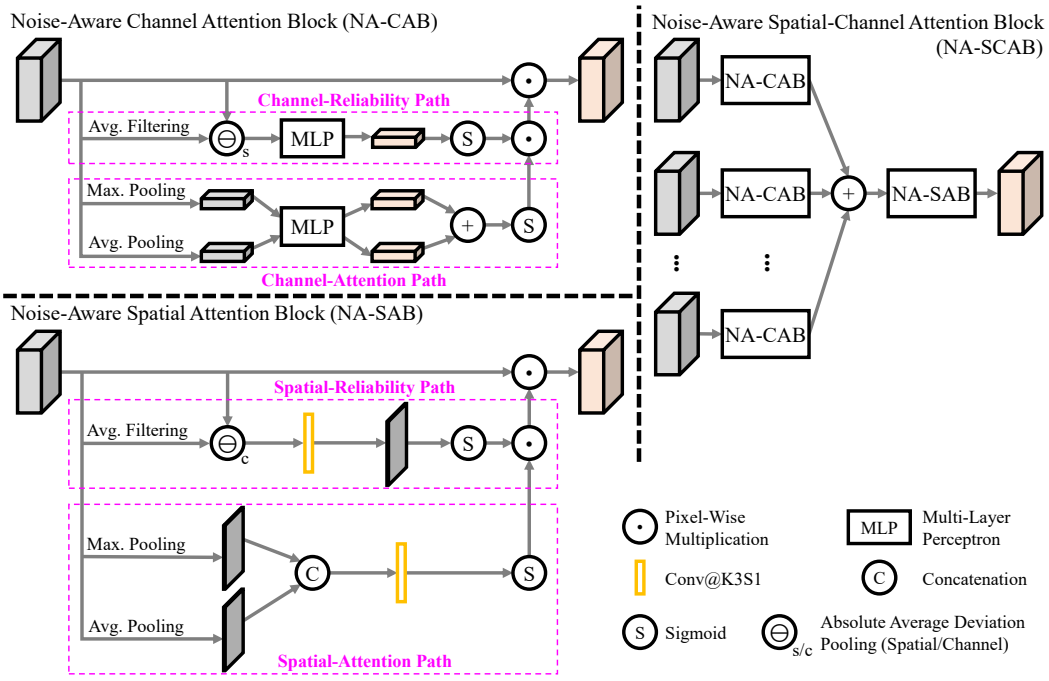


Figure 3. Architectural designs of the noise-aware spatial-channel attention block.

3.3.1. Noise-Aware Channel Attention

Given an input feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, NA-CAB estimates channel importance and channel reliability. The conventional channel-attention path uses global average pooling and global max pooling,

$$\mathbf{A}_c = \sigma(\mathrm{MLP}(\mathrm{GAP}(\mathbf{F})) + \mathrm{MLP}(\mathrm{GMP}(\mathbf{F}))), \tag{6}$$

where $\mathrm{GAP}(\cdot)$ and $\mathrm{GMP}(\cdot)$ denote global average pooling and global max pooling, respectively; $\mathrm{MLP}(\cdot)$ is a shared multilayer perceptron; and $\sigma(\cdot)$ is the sigmoid function.

To estimate channel reliability, a locally smoothed feature map is first obtained by average filtering:

$$\bar{\mathbf{F}} = \mathrm{AvgFilter}(\mathbf{F}). \tag{7}$$

The absolute deviation between $\mathbf{F}$ and $\bar{\mathbf{F}}$ captures unstable local responses:

$$\mathbf{R} = |\mathbf{F} - \bar{\mathbf{F}}|. \tag{8}$$

A channel-wise reliability vector is then computed as

$$\mathbf{Q}_c = \sigma(\mathrm{MLP}_r(\mathrm{AAP}_s(\mathbf{R}))), \tag{9}$$

where $\mathrm{AAP}_s(\cdot)$ denotes absolute average-deviation pooling over spatial locations. The final channel attention is

$$\hat{\mathbf{A}}_c = \mathbf{A}_c \odot \mathbf{Q}_c, \tag{10}$$

and the channel-refined feature is:

$$\mathbf{F}_c = \hat{\mathbf{A}}_c \odot \mathbf{F}. \tag{11}$$

A channel is thus emphasized only when it is both discriminative and reliable.

3.3.2. Noise-Aware Spatial Attention

NA-SAB estimates spatial importance and spatial reliability from $\mathbf{F}_c$. The spatial-attention path is defined as

$$\mathbf{A}_s = \sigma(\mathrm{Conv}_{3\times3}([\mathrm{AvgPool}_c(\mathbf{F}_c); \mathrm{MaxPool}_c(\mathbf{F}_c)])), \tag{12}$$

where $[\cdot;\cdot]$ denotes channel-wise concatenation.

The spatial-reliability path uses the local residual derived from Equation (8). Its spatial descriptor is computed by absolute average-deviation pooling across channels,

$$\mathbf{Q}_s = \sigma(\mathrm{Conv}_{3\times3}(\mathrm{AAP}_c(\mathbf{R}))), \tag{13}$$

where $\mathrm{AAP}_c(\cdot)$ denotes absolute average-deviation pooling over channels. The final spatial attention map is:

$$\hat{\mathbf{A}}_s = \mathbf{A}_s \odot \mathbf{Q}_s. \tag{14}$$

For a set of aligned input features $\{\mathbf{F}^{(m)}\}_{m=1}^{M}$ received by a branch, NA-SCAB first applies NA-CAB to each feature and combines the results:

$$\mathbf{F}_{\mathrm{sum}} = \sum_{m=1}^{M} \text{NA-CAB}\left(\mathbf{F}^{(m)}\right). \tag{15}$$

The final fused feature is obtained by:

$$\mathbf{F}_{\mathrm{out}} = \text{NA-SAB}(\mathbf{F}_{\mathrm{sum}}). \tag{16}$$

Unlike conventional spatial-channel attention, NA-SCAB considers feature reliability during both channel and spatial selection. It can therefore suppress noisy activations before they affect neighboring branches.

3.4. Evidence-Based Opinion Fusion

The final RDB output from each pseudo-view branch is converted into class evidence. Let $\mathbf{e}_j \in \mathbb{R}_+^K$ denote the nonnegative evidence vector produced by pseudo-view j, where K is the number of classes and $j \in \{1, \ldots, N\}$ ($N = 3$ for the design in Figure 1). A nonnegative activation function (ReLU in this study) is applied at the output layer of the final RDB to ensure that each element of $\mathbf{e}_j$ represents the evidence supporting a class.

The evidence vector is mapped to the parameters of a Dirichlet distribution

$$\boldsymbol{\alpha}_j = \mathbf{e}_j + \mathbf{1}, \tag{17}$$

where $\mathbf{1}$ is a K-dimensional vector of ones. The total Dirichlet strength is defined as:

$$S_j = \sum_{k=1}^{K} \alpha_{j,k}. \tag{18}$$

Based on subjective logic, the Dirichlet parameters are converted into an opinion

$$\boldsymbol{\omega}_j = \left(\mathbf{b}_j, u_j, \mathbf{a}\right), \tag{19}$$

where $\mathbf{b}_j = [b_{j,1}, \ldots, b_{j,K}]$ is the belief mass vector, u_j is the uncertainty mass, and $\mathbf{a}$ is the base-rate vector. The belief mass and uncertainty are computed as

$$b_{j,k} = \frac{e_{j,k}}{S_j}, \qquad u_j = \frac{K}{S_j}, \tag{20}$$

with a uniform base rate:

$$a_k = \frac{1}{K}. \tag{21}$$

The opinion satisfies the normalization condition:

$$\sum_{k=1}^{K} b_{j,k} + u_j = 1. \tag{22}$$

Thus, a pseudo-view with strong class-specific evidence produces high belief and low uncertainty. In contrast, weak or ambiguous evidence results in high uncertainty.

The opinions from all pseudo-views are fused using the subjective-logic averaging rule. The fused belief and uncertainty are obtained by:

$$\bar{b}_k = \frac{1}{N} \sum_{j=1}^{N} b_{j,k}, \qquad \bar{u} = \frac{1}{N} \sum_{j=1}^{N} u_j. \tag{23}$$

The fused opinion is expressed as:

$$\bar{\omega} = \left(\bar{\mathbf{b}}, \bar{u}, \mathbf{a}\right). \tag{24}$$

The corresponding expected class probability is:

$$p_k = \bar{b}_k + a_k \bar{u}. \tag{25}$$

The final prediction is determined from the class with the largest fused probability:

$$\hat{y} = \operatorname{argmax}_k p_k. \tag{26}$$

This evidence-based fusion preserves uncertainty information from individual pseudo-views. It also limits overconfident belief from a branch that produces weak or unreliable evidence, which is particularly important when cross-scale features are affected by noise.

3.5. Loss Optimization

Let $\mathbf{e}_j \in \mathbb{R}_+^K$ denote the evidence vector from pseudo-view j. As defined in Equation (17), the corresponding Dirichlet parameters are $\boldsymbol{\alpha}_j = \mathbf{e}_j + \mathbf{1}$, and the Dirichlet strength is given by Equation (18). For each pseudo-view, the expected cross-entropy loss is

$$\mathcal{L}_{\mathrm{CE}}^{j} = \sum_{k=1}^{K} y_k \Big[\psi(S_j) - \psi(\alpha_{j,k})\Big], \tag{27}$$

where $\mathbf{y}$ is the one-hot label vector and $\psi(\cdot)$ is the digamma function.

To discourage unsupported evidence and improve uncertainty calibration, a Kullback–Leibler (KL) regularization term is included,

$$\mathcal{L}_{\mathrm{KL}}^{j} = D_{\mathrm{KL}}\big[\mathrm{Dir}(\tilde{\boldsymbol{\alpha}}_j) \parallel \mathrm{Dir}(\mathbf{1})\big], \tag{28}$$

where

$$\tilde{\boldsymbol{\alpha}}_j = \mathbf{y} + (\mathbf{1} - \mathbf{y}) \odot \boldsymbol{\alpha}_j. \tag{29}$$

The total training objective sums the losses across all pseudo-views,

$$\mathcal{L} = \sum_{j=1}^{N} \left(\mathcal{L}_{\mathrm{CE}}^{j} + \lambda \mathcal{L}_{\mathrm{KL}}^{j} \right), \tag{30}$$

where λ balances classification learning and uncertainty regularization. This objective encourages every pseudo-view to produce discriminative evidence while limiting overconfident predictions from noisy or unreliable features.

4. Experimental Results

4.1. Experiment Settings

The proposed method is evaluated on the BreakHis [18], Oxford-IIIT Pet [19], and Chest X-ray [20] datasets. These datasets were selected because they cover distinct application domains, image characteristics, and classification challenges. BreakHis contains histopathological images with substantial intra-class variation across magnification levels. Oxford-IIIT Pet contains natural images with variations in pose, background, illumination, and appearance. Chest X-ray contains medical radiographs in which class-discriminative patterns can be subtle and affected by acquisition noise. Together, these datasets enable evaluation under diverse visual conditions and allow assessment of the proposed method in fine-grained natural-image and medical-image classification tasks.

BreakHis is a breast cancer histopathological image dataset containing 9109 images from 82 patients. The images are acquired at four magnification levels: 40×, 100×, 200×, and 400×. Each image is labeled as benign or malignant. The dataset includes variations in staining, tissue structure, cellular appearance, and image scale. Each magnification level can be regarded as an independent binary classification task.

Oxford-IIIT Pet contains 7349 natural images from 37 pet categories, including 25 dog breeds and 12 cat breeds. The images contain large variations in object pose, scale, viewpoint, illumination, background clutter, and occlusion. This dataset evaluates whether the proposed method can learn robust multiscale representations in fine-grained natural-image classification.

Chest X-ray contains frontal chest radiographs for pneumonia classification. The dataset includes 5863 images labeled as normal or pneumonia. Compared with color natural images, chest radiographs have limited texture and low contrast, while disease-related patterns can be localized and subtle. This dataset is used to evaluate the robustness of the proposed method when discriminative features are weak and susceptible to noise.

For all the datasets, five-fold cross-validation is used. The samples are divided into five mutually exclusive folds, and each fold is used once as the test set while the remaining four folds are used for training. The final performance is reported as the average over the five test folds. For BreakHis and Chest X-ray, the folds are created at the patient level to avoid data leakage. For Oxford-IIIT Pet, the folds are created at the image level.

Four metrics are used for evaluation: area under the receiver operating characteristic curve (AUC), accuracy (ACC), expected calibration error (ECE), and uncertainty-error correlation (ρ_{UE}). AUC and ACC measure classification performance, ECE measures the calibration of predicted probabilities, and ρ_{UE} quantifies the correlation between predictive uncertainty and classification error. A higher AUC or ACC indicates better classification performance, while a lower ECE indicates better calibration. A higher ρ_{UE} indicates that higher uncertainty is associated with a higher error rate, thereby supporting the reliability of uncertainty estimates.

ECE and ρ_{UE} are computed on the complete test set without excluding samples according to the uncertainty threshold. ECE is computed by partitioning the predicted

probabilities into 15 equal-width bins and calculating the weighted average of the absolute difference between accuracy and confidence in each bin. ρ_{UE} is computed as the Spearman correlation coefficient between the predictive uncertainty and the classification error. Test samples are grouped into 15 equal-width bins based on predictive uncertainty, and the average error rate is computed for each bin. The Spearman correlation between the average uncertainty and average error rate across bins is then computed.

The hyperparameters are selected using Microsoft Neural Network Intelligence (NNI) [21]. The AdamW optimizer [22] (Adam with decoupled weight decay) is used in which the learning rate is searched in the range $[10^{-4}, 10^{-1}]$ while the default weight-decay factor of 10^{-2} is used. λ is searched in the range $[0.5, 2]$, and the batch size is selected from $\{8, 32, 64, 256, 512, 1024\}$. Training is terminated early when the validation performance does not improve for ten consecutive epochs. The same cross-validation protocol and hyperparameter-search procedure are applied to all the compared methods to ensure a fair evaluation.

4.2. Comparison with Pseudo-Multiview Learning Series

The proposed method is compared with conventional PML and stability-enhanced PML (SE-PML). Figure 4 illustrates the PML and SE-PML architectures used in this comparison. To ensure a fair evaluation, all the methods use the same number of pseudo-views and comparable intermediate feature sizes. All the input images are resized to 224×224. The number of pseudo-views is set to two, three, and four. PML generates pseudo-views using independent branches, whereas SE-PML generates lower-resolution pseudo-views through downsampling and top-down feature propagation. The proposed method follows the multiscale design of SE-PML but replaces unidirectional propagation with dual-sampler bidirectional feature flow and noise-aware spatial-channel attention.

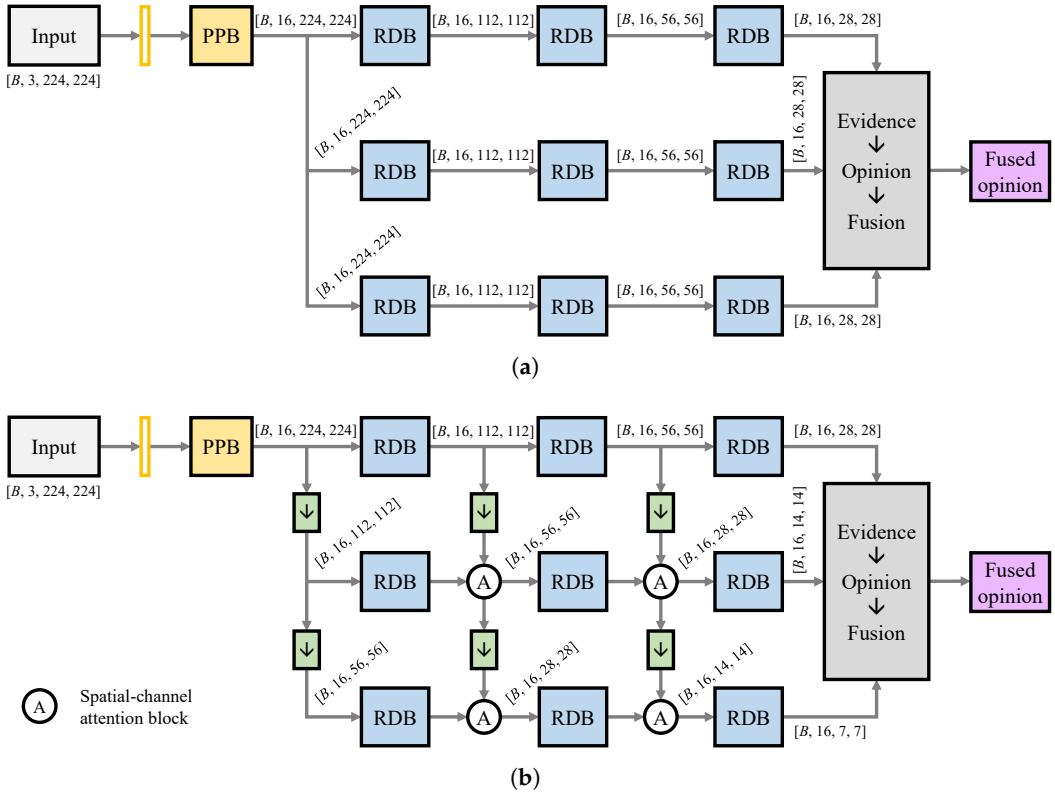


Figure 4. Overall architectural design of baseline methods for three pseudo-views. (**a**) Pseudo-multiview learning; (**b**) stability-enhanced pseudo-multiview learning.

Table 1 reports the AUC, ACC, ECE, and ρ_{UE} results. The proposed method achieves the best AUC and ACC in all the evaluation settings. Averaged over these settings, it improves AUC and ACC by 0.060 and 0.074, respectively, over SE-PML and by 0.116 and 0.135 over conventional PML. The proposed method also generally produces lower ECE and higher ρ_{UE}, indicating that its predictive probabilities are better calibrated and its uncertainty estimates are more closely associated with classification errors.

The performance gap is particularly clear when four pseudo-views are used. Conventional PML shows a marked performance drop as the number of views increases. For example, on Oxford-IIIT Pet, its AUC decreases from 0.829 with three views to 0.688 with four views, and its ACC decreases from 0.744 to 0.609. Similar decreases are observed on Chest X-ray, where the AUC and ACC fall to 0.683 and 0.599, respectively. This behavior reflects the optimization difficulty and computational burden caused by additional independent pseudo-view branches, as pointed out in a previous study [8]. In contrast, SE-PML and the proposed method maintain or improve their performance by sharing multiscale representations. With four views, the proposed method obtains an average improvement of 0.061 in AUC and 0.072 in ACC over SE-PML.

Table 1. Performance comparison between the proposed method and pseudo-multiview learning series. The best results are boldfaced.

Dataset	Number of Views	AUC			ACC			ECE			ρ_{UE}		
		PML	SE-PML	Proposed	PML	SE-PML	Proposed	PML	SE-PML	Proposed	PML	SE-PML	Proposed
BreakHis 40×	2	0.590 ± 0.039	0.618 ± 0.037	**0.701 ± 0.033**	0.622 ± 0.040	0.645 ± 0.038	**0.727 ± 0.036**	0.066 ± 0.013	0.047 ± 0.009	**0.025 ± 0.005**	0.844 ± 0.071	0.901 ± 0.071	**0.929 ± 0.087**
	3	0.611 ± 0.041	0.638 ± 0.037	**0.712 ± 0.035**	0.634 ± 0.042	0.662 ± 0.036	**0.740 ± 0.035**	0.069 ± 0.011	0.048 ± 0.010	**0.027 ± 0.006**	**0.894 ± 0.073**	0.840 ± 0.111	0.827 ± 0.040
	4	0.565 ± 0.048	0.654 ± 0.035	**0.735 ± 0.040**	0.598 ± 0.049	0.676 ± 0.034	**0.749 ± 0.038**	0.070 ± 0.013	0.048 ± 0.012	**0.027 ± 0.005**	0.824 ± 0.103	0.759 ± 0.153	**0.870 ± 0.080**
BreakHis 100×	2	0.652 ± 0.047	0.672 ± 0.035	**0.754 ± 0.032**	0.631 ± 0.072	0.651 ± 0.058	**0.743 ± 0.044**	0.081 ± 0.015	0.051 ± 0.010	**0.030 ± 0.006**	0.873 ± 0.058	0.768 ± 0.035	**0.946 ± 0.061**
	3	0.669 ± 0.044	0.685 ± 0.033	**0.766 ± 0.028**	0.645 ± 0.067	0.668 ± 0.054	**0.755 ± 0.049**	0.079 ± 0.014	0.050 ± 0.009	**0.029 ± 0.007**	0.718 ± 0.031	0.741 ± 0.074	**0.883 ± 0.076**
	4	0.621 ± 0.051	0.701 ± 0.032	**0.781 ± 0.030**	0.602 ± 0.073	0.683 ± 0.049	**0.772 ± 0.047**	0.082 ± 0.014	0.052 ± 0.009	**0.031 ± 0.006**	0.801 ± 0.023	0.774 ± 0.064	**0.915 ± 0.172**
BreakHis 200×	2	0.688 ± 0.027	0.707 ± 0.031	**0.788 ± 0.032**	0.641 ± 0.053	0.668 ± 0.045	**0.781 ± 0.040**	0.075 ± 0.009	0.046 ± 0.010	**0.025 ± 0.004**	0.765 ± 0.079	0.779 ± 0.180	**0.859 ± 0.027**
	3	0.703 ± 0.025	0.723 ± 0.028	**0.796 ± 0.028**	0.654 ± 0.049	0.684 ± 0.042	**0.785 ± 0.039**	0.077 ± 0.009	0.047 ± 0.011	**0.026 ± 0.004**	0.817 ± 0.068	0.877 ± 0.143	**0.952 ± 0.053**
	4	0.656 ± 0.036	0.739 ± 0.026	**0.807 ± 0.028**	0.612 ± 0.057	0.701 ± 0.040	**0.790 ± 0.038**	0.075 ± 0.010	0.046 ± 0.009	**0.026 ± 0.005**	0.804 ± 0.107	0.806 ± 0.161	**0.971 ± 0.131**
BreakHis 400×	2	0.690 ± 0.032	0.711 ± 0.029	**0.791 ± 0.030**	0.635 ± 0.029	0.662 ± 0.027	**0.779 ± 0.035**	0.079 ± 0.012	0.049 ± 0.008	**0.028 ± 0.003**	0.827 ± 0.044	0.819 ± 0.039	**0.922 ± 0.139**
	3	0.702 ± 0.030	0.726 ± 0.027	**0.801 ± 0.028**	0.648 ± 0.031	0.676 ± 0.026	**0.782 ± 0.033**	0.080 ± 0.014	0.049 ± 0.009	**0.029 ± 0.004**	0.783 ± 0.112	0.792 ± 0.163	**0.955 ± 0.113**
	4	0.655 ± 0.038	0.741 ± 0.025	**0.819 ± 0.022**	0.603 ± 0.038	0.694 ± 0.023	**0.800 ± 0.036**	0.081 ± 0.012	0.050 ± 0.009	**0.030 ± 0.004**	**0.814 ± 0.088**	0.781 ± 0.048	0.812 ± 0.092
Oxford-IIIT Pet	2	0.818 ± 0.022	0.832 ± 0.028	**0.862 ± 0.017**	0.733 ± 0.029	0.751 ± 0.030	**0.797 ± 0.026**	0.150 ± 0.048	0.088 ± 0.015	**0.052 ± 0.009**	0.829 ± 0.090	0.888 ± 0.125	**0.899 ± 0.111**
	3	0.829 ± 0.020	0.845 ± 0.026	**0.880 ± 0.023**	0.744 ± 0.027	0.764 ± 0.030	**0.819 ± 0.027**	0.152 ± 0.050	0.092 ± 0.017	**0.055 ± 0.008**	0.855 ± 0.075	0.800 ± 0.093	**0.864 ± 0.064**
	4	0.688 ± 0.047	0.861 ± 0.025	**0.902 ± 0.023**	0.609 ± 0.033	0.778 ± 0.027	**0.833 ± 0.020**	0.155 ± 0.051	0.089 ± 0.017	**0.053 ± 0.010**	0.828 ± 0.160	0.844 ± 0.098	**0.961 ± 0.134**
Chest X-ray	2	0.917 ± 0.010	0.930 ± 0.012	**0.932 ± 0.013**	0.869 ± 0.012	0.881 ± 0.013	**0.892 ± 0.012**	0.171 ± 0.063	0.102 ± 0.024	**0.081 ± 0.012**	0.784 ± 0.056	0.780 ± 0.084	**0.902 ± 0.051**
	3	0.924 ± 0.009	0.938 ± 0.010	**0.951 ± 0.009**	0.875 ± 0.011	0.892 ± 0.010	**0.910 ± 0.013**	0.174 ± 0.052	0.108 ± 0.027	**0.083 ± 0.014**	0.756 ± 0.019	0.743 ± 0.067	**0.864 ± 0.174**
	4	0.683 ± 0.015	0.945 ± 0.018	**0.962 ± 0.011**	0.599 ± 0.017	0.904 ± 0.012	**0.922 ± 0.010**	0.168 ± 0.055	0.105 ± 0.030	**0.083 ± 0.013**	0.772 ± 0.080	0.754 ± 0.149	**0.867 ± 0.099**

The improvements are not uniform across the datasets, reflecting differences in image content and task difficulty. The largest gains over SE-PML are observed on BreakHis at all four magnifications. With four views, the proposed method improves AUC/ACC over SE-PML from 0.654/0.676 to 0.735/0.749 on BreakHis 40× and from 0.701/0.683 to 0.781/0.772 on BreakHis 100×. At these magnifications, discriminative information is distributed across cellular details, tissue texture, and broader morphological structures. The bidirectional exchange between high- and low-resolution branches therefore enables contextual features to guide detailed representations while allowing fine-grained information to refine the coarser branches. This complementary interaction is particularly beneficial when the relevant visual cues occur at multiple scales.

The gains are smaller on Chest X-ray, although the proposed method remains consistently superior. With four views, it achieves an AUC of 0.962 and an ACC of 0.922 compared with 0.945 and 0.904 for SE-PML. The smaller relative improvement is likely because chest radiographs have lower texture complexity and fewer distinct scale-specific patterns than histopathological images. Pneumonia-related abnormalities can also appear as diffuse low-contrast regions, so the simpler multiscale representation of SE-PML already captures much of the available discriminative information. Nevertheless, the proposed bidirectional fusion remains useful for integrating localized and global radiographic cues and suppressing acquisition-related noise. Thus, the method is the most advantageous for

tasks with heterogeneous, fine-grained, and multiscale discriminative structures while still providing consistent benefits in lower-texture medical-image classification.

The calibration and uncertainty results support this interpretation. For example, in the four-view setting, the proposed method reduces ECE from 0.048 to 0.027 on BreakHis 40×, from 0.089 to 0.053 on Oxford-IIIT Pet, and from 0.105 to 0.083 on Chest X-ray relative to SE-PML. The corresponding ρ_{UE} values are 0.870, 0.961, and 0.867, respectively, showing that higher estimated uncertainty generally corresponds to a higher error rate. These results indicate that the proposed method improves not only discriminative performance but also the reliability of its uncertainty estimates.

Figure 5 compares the convergence behavior of PML, SE-PML, and the proposed method on six datasets. The proposed method converges faster and reaches a higher accuracy than both baselines in every case. On the four BreakHis subsets and Oxford-IIIT Pet, the proposed method shows a steep accuracy increase during the early training stage and stabilizes within approximately 10-20 epochs. In contrast, PML converges more slowly and remains at the lowest accuracy level, while SE-PML provides a moderate improvement through multiscale feature extraction.

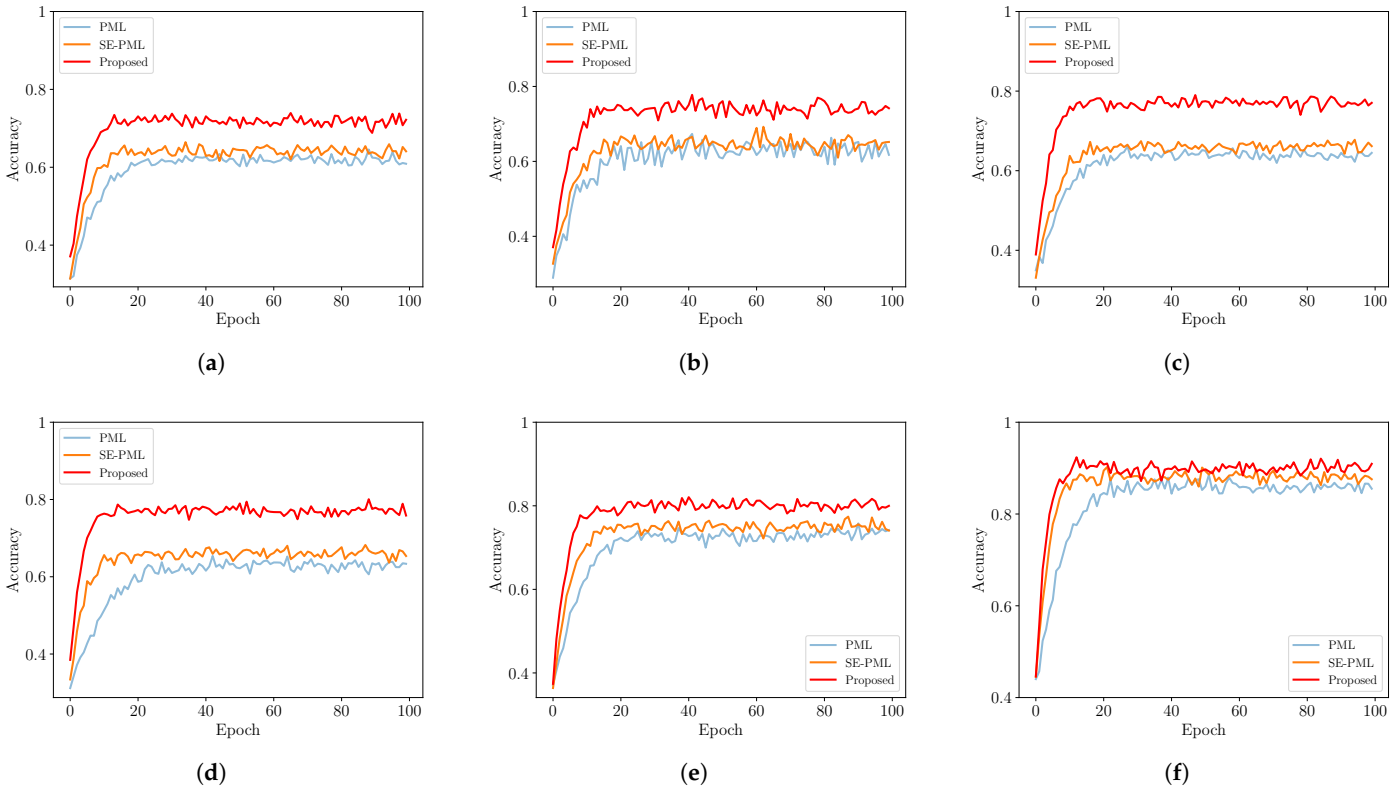


Figure 5. Model convergence on different real-world datasets. (**a**) BreakHis 40×; (**b**) BreakHis 100×; (**c**) BreakHis 200×; (**d**) BreakHis 400×; (**e**) Oxford-IIIT Pet; (**f**) Chest X-ray.

The difference is the most evident on BreakHis at 40×, 100×, 200×, and 400×, where the proposed method maintains a clear accuracy margin after convergence. This result indicates that bidirectional feature exchange allows lower-resolution branches to provide stable contextual information during the optimization of higher-resolution branches. The noise-aware spatial-channel attention further limits the influence of unreliable cross-scale features, resulting in more stable accuracy curves after the initial convergence period.

On Chest X-ray, all the methods converge rapidly and achieve relatively high accuracy because the task is less challenging than the fine-grained BreakHis and Oxford-IIIT Pet settings. Nevertheless, the proposed method still reaches the highest final accuracy and

exhibits a stable training trajectory. Overall, Figure 5 supports the quantitative results in Table 1 by showing that the proposed method improves not only final classification accuracy but also convergence speed and optimization stability.

4.3. Comparison with Uncertainty-Aware Methods

The proposed method is compared with three single-view uncertainty-aware methods: Monte Carlo dropout (MCDO) [23], uncertainty-aware attention (UA) [24], and evidential deep learning (EDL) [25]. PML and SE-PML are also included as multiview uncertainty-aware baselines. For a fair comparison, MCDO, UA, and EDL are implemented using the first branch of the proposed architecture. PML, SE-PML, and the proposed method use two pseudo-views. Thus, the comparison evaluates the benefit of pseudo-view learning and the effectiveness of the proposed bidirectional feature flow and noise-aware attention.

Table 2 reports the AUC, ACC, ECE, and ρ_{UE} results. The proposed method achieves the best AUC and ACC in all six evaluation settings. Averaged across the six settings, it obtains an AUC of 0.805 and an ACC of 0.787 compared with 0.745 and 0.710 for the strongest baseline, SE-PML. Thus, the proposed method improves AUC and ACC by 0.060 and 0.077, respectively. It also provides the lowest ECE in all six settings, an average ECE of 0.040 compared with 0.064 for SE-PML. This indicates that the proposed method produces more reliable predictive probabilities in addition to improving classification performance.

Table 2. Performance comparison between uncertainty-aware approaches and the proposed method. The best results are boldfaced.

Metric	Dataset		MCDO	UA	EDL	PML	SE-PML	Proposed
AUC	BreakHis	40×	0.567 ± 0.033	0.582 ± 0.029	0.585 ± 0.028	0.590 ± 0.039	0.618 ± 0.037	**0.701 ± 0.033**
		100×	0.604 ± 0.030	0.626 ± 0.028	0.637 ± 0.026	0.652 ± 0.047	0.672 ± 0.035	**0.754 ± 0.032**
		200×	0.671 ± 0.027	0.687 ± 0.023	0.694 ± 0.021	0.688 ± 0.027	0.707 ± 0.031	**0.788 ± 0.032**
		400×	0.650 ± 0.034	0.671 ± 0.032	0.683 ± 0.030	0.690 ± 0.032	0.711 ± 0.029	**0.791 ± 0.030**
	Oxford-IIIT Pet		0.781 ± 0.025	0.801 ± 0.020	0.808 ± 0.019	0.818 ± 0.022	0.832 ± 0.028	**0.862 ± 0.017**
	Chest X-ray		0.891 ± 0.011	0.905 ± 0.009	0.913 ± 0.009	0.917 ± 0.010	0.930 ± 0.012	**0.932 ± 0.013**
ACC	BreakHis	40×	0.586 ± 0.027	0.605 ± 0.024	0.616 ± 0.022	0.622 ± 0.040	0.645 ± 0.038	**0.727 ± 0.036**
		100×	0.597 ± 0.026	0.612 ± 0.024	0.625 ± 0.025	0.631 ± 0.072	0.651 ± 0.058	**0.743 ± 0.044**
		200×	0.602 ± 0.025	0.620 ± 0.022	0.631 ± 0.021	0.641 ± 0.053	0.668 ± 0.045	**0.781 ± 0.040**
		400×	0.599 ± 0.028	0.616 ± 0.025	0.629 ± 0.023	0.635 ± 0.029	0.662 ± 0.027	**0.779 ± 0.035**
	Oxford-IIIT Pet		0.688 ± 0.025	0.712 ± 0.020	0.720 ± 0.018	0.733 ± 0.029	0.751 ± 0.030	**0.797 ± 0.026**
	Chest X-ray		0.833 ± 0.015	0.850 ± 0.014	0.861 ± 0.013	0.869 ± 0.012	0.881 ± 0.013	**0.892 ± 0.012**
ECE	BreakHis	40×	0.101 ± 0.013	0.075 ± 0.020	0.053 ± 0.008	0.066 ± 0.013	0.047 ± 0.009	**0.025 ± 0.005**
		100×	0.126 ± 0.015	0.081 ± 0.026	0.074 ± 0.012	0.081 ± 0.015	0.051 ± 0.010	**0.030 ± 0.006**
		200×	0.137 ± 0.022	0.083 ± 0.017	0.077 ± 0.011	0.075 ± 0.009	0.046 ± 0.010	**0.025 ± 0.004**
		400×	0.144 ± 0.020	0.086 ± 0.030	0.072 ± 0.009	0.079 ± 0.012	0.049 ± 0.008	**0.028 ± 0.003**
	Oxford-IIIT Pet		0.198 ± 0.032	0.152 ± 0.041	0.138 ± 0.035	0.150 ± 0.048	0.088 ± 0.015	**0.052 ± 0.009**
	Chest X-ray		0.226 ± 0.055	0.168 ± 0.059	0.122 ± 0.029	0.171 ± 0.063	0.102 ± 0.024	**0.081 ± 0.012**
ρ_{UE}	BreakHis	40×	0.782 ± 0.068	0.739 ± 0.077	0.850 ± 0.034	0.844 ± 0.071	0.901 ± 0.071	**0.929 ± 0.087**
		100×	0.785 ± 0.101	0.749 ± 0.092	0.794 ± 0.045	0.873 ± 0.058	0.768 ± 0.035	**0.946 ± 0.061**
		200×	0.827 ± 0.077	0.763 ± 0.056	0.776 ± 0.043	0.765 ± 0.079	0.779 ± 0.180	**0.859 ± 0.027**
		400×	0.761 ± 0.066	0.776 ± 0.129	0.825 ± 0.068	0.827 ± 0.044	0.819 ± 0.039	**0.922 ± 0.139**
	Oxford-IIIT Pet		0.784 ± 0.110	0.723 ± 0.071	0.866 ± 0.108	0.829 ± 0.090	0.888 ± 0.125	**0.899 ± 0.111**
	Chest X-ray		0.786 ± 0.097	0.792 ± 0.092	**0.910 ± 0.104**	0.784 ± 0.056	0.780 ± 0.084	0.902 ± 0.051

The proposed method also generally provides the strongest relationship between uncertainty and prediction errors. Its average ρ_{UE} is 0.910 compared with 0.823 for SE-PML. In particular, the proposed method achieves ρ_{UE} values of 0.929, 0.946, 0.859, 0.922, 0.899, and 0.902 across the six settings. These results indicate that samples assigned higher uncertainty tend to have higher error rates, supporting the usefulness of the estimated uncertainty. The only exception is Chest X-ray, where EDL obtains a slightly higher correlation (0.910 versus 0.902 for the proposed method).

The single-view methods show lower discriminative performance than the pseudo-multiview methods. Among them, EDL performs the best, demonstrating average AUC

and ACC values of 0.720 and 0.680, respectively. However, it remains below PML, SE-PML, and the proposed method. EDL also provides competitive uncertainty estimates in some settings, including the highest ρ_{UE} on Chest X-ray (0.910), but its overall classification performance remains lower. These results indicate that uncertainty estimation alone is insufficient to capture the complementary information provided by multiple feature representations. MCDO and UA show further performance reductions, presumably because they rely on a single feature stream and cannot exploit cross-scale feature interactions.

The advantage of the proposed method is the most evident on the BreakHis subsets. Compared with SE-PML, it improves AUC by 0.083, 0.082, 0.081, and 0.080 at 40×, 100×, 200×, and 400× magnifications, respectively. The corresponding ACC improvements are 0.082, 0.092, 0.113, and 0.117. It also substantially reduces ECE from 0.047, 0.051, 0.046, and 0.049 for SE-PML to 0.025, 0.030, 0.025, and 0.028, respectively. The corresponding ρ_{UE} values increase from 0.901, 0.768, 0.779, and 0.819 to 0.929, 0.946, 0.859, and 0.922. These results show that the proposed method is particularly effective for histopathological images, where discriminative tissue patterns occur at multiple scales.

On Oxford-IIIT Pet, the proposed method achieves an AUC of 0.862 and an ACC of 0.797, improving on SE-PML by 0.030 and 0.046, respectively. It also reduces ECE from 0.088 to 0.052 and increases ρ_{UE} from 0.888 to 0.899. On Chest X-ray, the proposed method achieves the highest AUC (0.932) and ACC (0.892), although the AUC gain over SE-PML is only 0.002. This limited AUC improvement is expected because SE-PML already performs strongly on this dataset. Nevertheless, the proposed method improves ACC by 0.011, reduces ECE from 0.102 to 0.081, and increases ρ_{UE} from 0.780 to 0.902. Thus, its primary benefit on Chest X-ray is improved calibration and uncertainty quality rather than a large gain in ranking performance.

These results indicate that the proposed framework combines the benefits of multiview evidence fusion and uncertainty-aware learning. The dual-sampler module allows low- and high-resolution branches to exchange complementary features, while the noise-aware spatial-channel attention reduces the influence of unreliable information during cross-scale fusion. Consequently, the proposed method improves not only AUC and ACC but also the calibration and error-awareness of its predictions.

4.4. Comparison with Multiscale Feature Fusion Methods

Table 3 compares the proposed method with the multiscale feature-fusion baselines: PANet [11] and BiFPN [12]. Because these methods do not explicitly model predictive uncertainty, Monte Carlo dropout [23] is applied during inference to obtain uncertainty estimates for computing ECE and ρ_{UE}.

The proposed method and BiFPN achieve comparable AUC and ACC. On BreakHis, the proposed method obtains AUC/ACC values of 0.701/0.727, 0.754/0.743, 0.788/0.781, and 0.791/0.779 at 40×, 100×, 200×, and 400×, respectively. On Oxford-IIIT Pet and Chest X-ray, it achieves AUC/ACC values of 0.862/0.797 and 0.932/0.892. These results indicate that the proposed dual-sampler bidirectional feature flow is competitive with well-known multiscale fusion methods in integrating complementary information across resolutions.

The advantage is particularly evident in calibration and uncertainty quality. The proposed method produces the lowest ECE in every setting, ranging from 0.025 to 0.081, whereas the corresponding ECE values for PANet and BiFPN range from 0.075 to 0.145 and from 0.054 to 0.140, respectively. The proposed method also achieves the highest ρ_{UE} in every setting. For example, on BreakHis at 100×, ρ_{UE} increases to 0.946 compared with 0.745 for PANet and 0.843 for BiFPN. On Chest X-ray, the proposed method obtains 0.902, whereas PANet and BiFPN obtain 0.834 and 0.869, respectively. Thus, higher estimated uncertainty from the proposed method is more consistently associated with prediction errors.

Although Monte Carlo dropout provides uncertainty estimates for PANet and BiFPN, it is an inference-time approximation rather than an uncertainty representation learned from class evidence. In contrast, the proposed method derives uncertainty directly from the fused evidential representation. The results therefore suggest that the proposed approach improves not only discriminative performance but also the calibration and error-awareness of its uncertainty estimates. Overall, the comparison demonstrates the benefit of combining bidirectional multiscale interaction with explicit evidential uncertainty modeling.

Table 3. Performance comparison between the proposed method and multiscale feature-fusion approaches. The best results are boldfaced.

Dataset		AUC			ACC			ECE			ρ_{UE}		
		PANet	BiFPN	Proposed	PANet	BiFPN	Proposed	PANet	BiFPN	Proposed	PANet	BiFPN	Proposed
BreakHis	40×	0.692 ± 0.049	**0.718 ± 0.034**	0.701 ± 0.033	0.675 ± 0.044	0.714 ± 0.049	**0.727 ± 0.036**	0.075 ± 0.015	0.054 ± 0.010	**0.025 ± 0.005**	0.768 ± 0.075	0.853 ± 0.062	**0.929 ± 0.087**
	100×	0.737 ± 0.045	0.724 ± 0.033	**0.754 ± 0.032**	0.704 ± 0.042	0.723 ± 0.044	**0.743 ± 0.044**	0.088 ± 0.012	0.062 ± 0.014	**0.030 ± 0.006**	0.745 ± 0.055	0.843 ± 0.069	**0.946 ± 0.061**
	200×	0.746 ± 0.044	0.729 ± 0.025	**0.788 ± 0.032**	0.779 ± 0.039	**0.782 ± 0.027**	0.781 ± 0.040	0.112 ± 0.011	0.124 ± 0.013	**0.025 ± 0.004**	0.712 ± 0.063	0.770 ± 0.029	**0.859 ± 0.027**
	400×	0.801 ± 0.036	**0.814 ± 0.020**	0.791 ± 0.030	0.783 ± 0.035	**0.784 ± 0.026**	0.779 ± 0.035	0.117 ± 0.012	0.129 ± 0.012	**0.028 ± 0.003**	0.739 ± 0.050	0.866 ± 0.083	**0.922 ± 0.139**
Oxford-IIIT Pet		0.846 ± 0.028	0.858 ± 0.019	**0.862 ± 0.017**	0.793 ± 0.027	**0.839 ± 0.016**	0.797 ± 0.026	0.135 ± 0.009	0.129 ± 0.010	**0.052 ± 0.009**	0.848 ± 0.117	0.895 ± 0.102	**0.899 ± 0.111**
Chest X-ray		0.929 ± 0.023	0.919 ± 0.017	**0.932 ± 0.013**	0.830 ± 0.022	0.846 ± 0.014	**0.892 ± 0.012**	0.145 ± 0.014	0.140 ± 0.007	**0.081 ± 0.012**	0.834 ± 0.078	0.869 ± 0.053	**0.902 ± 0.051**

4.5. Exclusion Rate

Table 4 reports the exclusion rates obtained with a fixed uncertainty threshold of 0.5 for all the methods. The results show that the proposed method generally achieves a favorable balance between prediction coverage and uncertainty-based rejection. On BreakHis, it produces the lowest exclusion rate at 40×, 100×, 200×, and 400× magnification with rates of 15.81%, 18.24%, 20.74%, and 19.41%, respectively. On Oxford-IIIT Pet, its exclusion rate is 10.64%, which is comparable to the lowest values (10.63% and 10.60%) obtained by EDL and BiFPN. These results indicate that the proposed method can retain most samples while identifying only a small fraction as highly uncertain.

Table 4. Exclusion rates (count followed by % in brackets) of the proposed method and benchmarking approaches. The best results are boldfaced.

Dataset		MCDO	UA	EDL	PANet	BiFPN	PML	SE-PML	Proposed
BreakHis	40×	3.20 ± 0.84 (19.56 ± 5.18)	3.20 ± 0.84 (19.49 ± 4.99)	2.80 ± 0.45 (17.06 ± 2.61)	3.40 ± 0.55 (20.74 ± 3.29)	3.00 ± 1.00 (18.31 ± 6.10)	3.40 ± 0.55 (20.74 ± 3.29)	3.00 ± 0.71 (18.24 ± 3.93)	**2.60 ± 0.89 (15.81 ± 5.16)**
	100×	3.60 ± 1.14 (21.91 ± 6.75)	3.80 ± 0.84 (23.09 ± 4.51)	3.20 ± 0.84 (19.41 ± 4.54)	3.40 ± 1.14 (20.66 ± 6.61)	3.40 ± 1.34 (20.59 ± 7.69)	3.60 ± 0.55 (21.91 ± 2.95)	3.40 ± 0.89 (20.66 ± 5.14)	**3.00 ± 0.71 (18.24 ± 3.93)**
	200×	4.20 ± 0.84 (25.51 ± 4.38)	4.00 ± 0.71 (24.34 ± 3.82)	3.60 ± 0.55 (21.99 ± 3.53)	4.00 ± 0.71 (24.34 ± 3.82)	4.00 ± 1.00 (24.26 ± 5.35)	3.60 ± 0.55 (21.99 ± 3.53)	3.60 ± 0.55 (21.91 ± 2.95)	**3.40 ± 0.55 (20.74 ± 3.29)**
	400×	4.40 ± 0.55 (26.84 ± 3.31)	4.60 ± 0.55 (28.09 ± 3.61)	3.80 ± 0.45 (23.16 ± 2.57)	4.20 ± 0.84 (25.66 ± 5.25)	4.20 ± 0.84 (25.51 ± 4.38)	3.80 ± 0.45 (23.16 ± 2.57)	3.80 ± 0.45 (23.16 ± 2.57)	**3.20 ± 0.84 (19.41 ± 4.54)**
Oxford-IIIT Pet		174.60 ± 16.46 (11.88 ± 1.12)	182.40 ± 18.58 (12.41 ± 1.26)	156.20 ± 14.72 (10.63 ± 1.00)	160.80 ± 17.70 (10.94 ± 1.20)	**155.80 ± 15.09 (10.60 ± 1.02)**	165.80 ± 9.28 (11.28 ± 0.63)	162.60 ± 8.62 (11.06 ± 0.59)	156.40 ± 6.66 (10.64 ± 0.45)
Chest X-ray		147.00 ± 5.70 (12.54 ± 0.49)	151.60 ± 10.16 (12.93 ± 0.86)	153.40 ± 6.66 (13.08 ± 0.56)	136.80 ± 14.48 (11.67 ± 1.24)	137.00 ± 14.35 (11.68 ± 1.23)	144.40 ± 10.16 (12.31 ± 0.87)	138.00 ± 11.73 (11.77 ± 1.00)	**125.60 ± 11.33 (10.71 ± 0.97)**

The exclusion rate on Oxford-IIIT Pet is comparable to the lowest values despite the greater variation in pose, scale, viewpoint, illumination, background, and occlusion in this dataset. This observation indicates that the proposed method remains selective on visually ambiguous images without excluding an excessive number of samples.

On Chest X-ray, the proposed method excludes approximately 126 samples, which is fewer than the other compared methods shown in the table. Despite retaining more samples, it achieves the best classification performance and strong calibration and uncertainty-error correlation. This result suggests that the proposed method is sufficiently discriminative to assign low uncertainty to a larger number of correctly handled radiographs while still identifying difficult cases.

Overall, the exclusion results support the effectiveness of the proposed method at the common threshold of 0.5. Its lowest exclusion rates among the compared methods on BreakHis and Chest X-ray, together with a comparable rate to the lowest on Oxford-IIIT Pet, indicate high prediction coverage even under substantial visual ambiguity. Combined with the AUC, ACC, ECE, and uncertainty-error correlation reported in the preceding experiments, these results show that the proposed method provides a balance between coverage and selective rejection.

4.6. Ablation Study

4.6.1. Bidirectional Feature Flow and Noise-Aware Attention

Table 5 evaluates the individual contributions of bidirectional feature flow and noise-aware attention. "w/o Att" removes the channel- and spatial-reliability paths in the NA-SCAB (shown in Figure 3). Conversely, "w/o Bidir" replaces the bidirectional feature flow with the corresponding top-down configuration. The complete model achieves the best AUC and ACC on all six evaluation settings. For example, on BreakHis at 40×, the complete model improves AUC from 0.683 and 0.634 to 0.701 and ACC from 0.714 and 0.653 to 0.727 compared with the w/o Att and w/o Bidir variants, respectively. Similar improvements are observed on Oxford-IIIT Pet and Chest X-ray, where the complete model achieves AUC/ACC values of 0.862/0.797 and 0.932/0.892, respectively.

Table 5. Ablation study on the contribution of bidirectional feature flow and noise-aware attention. "w/o Att" denotes the proposed method without noise-aware attention. "w/o Bidir" denotes the proposed method without bidirectional feature flow. The best results are boldfaced.

Dataset		AUC			ACC			ECE			ρ_{UE}		
		w/o Att	w/o Bidir	Full	w/o Att	w/o Bidir	Full	w/o Att	w/o Bidir	Full	w/o Att	w/o Bidir	Full
BreakHis	40×	0.683 ± 0.034	0.634 ± 0.033	**0.701 ± 0.033**	0.714 ± 0.038	0.653 ± 0.040	**0.727 ± 0.036**	0.029 ± 0.007	0.046 ± 0.010	**0.025 ± 0.005**	0.912 ± 0.100	0.898 ± 0.075	**0.929 ± 0.087**
	100×	0.744 ± 0.032	0.686 ± 0.030	**0.754 ± 0.032**	0.730 ± 0.043	0.666 ± 0.047	**0.743 ± 0.044**	0.035 ± 0.008	0.048 ± 0.010	**0.030 ± 0.006**	0.927 ± 0.074	0.770 ± 0.046	**0.946 ± 0.061**
	200×	0.772 ± 0.022	0.720 ± 0.027	**0.788 ± 0.032**	0.761 ± 0.036	0.695 ± 0.044	**0.781 ± 0.040**	0.032 ± 0.007	0.045 ± 0.011	**0.025 ± 0.004**	**0.872 ± 0.058**	0.801 ± 0.152	0.859 ± 0.027
	400×	0.781 ± 0.025	0.727 ± 0.026	**0.791 ± 0.030**	0.769 ± 0.030	0.680 ± 0.036	**0.779 ± 0.035**	0.035 ± 0.005	0.051 ± 0.009	**0.028 ± 0.003**	0.905 ± 0.126	0.826 ± 0.044	**0.922 ± 0.139**
Oxford-IIIT Pet		0.849 ± 0.019	0.853 ± 0.020	**0.862 ± 0.017**	0.784 ± 0.028	0.756 ± 0.025	**0.797 ± 0.026**	0.060 ± 0.011	0.074 ± 0.012	**0.052 ± 0.009**	0.893 ± 0.091	0.882 ± 0.127	**0.899 ± 0.111**
Chest X-ray		0.910 ± 0.015	0.887 ± 0.016	**0.932 ± 0.013**	0.881 ± 0.015	0.848 ± 0.018	**0.892 ± 0.012**	0.089 ± 0.010	0.092 ± 0.011	**0.081 ± 0.012**	0.864 ± 0.073	0.795 ± 0.091	**0.902 ± 0.051**

The results also show that the complete model provides the lowest ECE in all six settings, indicating better calibration of the fused predictive probabilities. The improvement is particularly clear on Chest X-ray, where ECE decreases from 0.089 and 0.092 for the two ablated variants to 0.081 for the complete model. This result suggests that bidirectional feature exchange and noise-aware attention improve not only classification performance but also the reliability of the predicted probabilities.

The Spearman correlation between uncertainty and error rate is also generally the highest for the complete model. It reaches 0.929, 0.946, 0.922, 0.899, and 0.902 on BreakHis at 40×, 100×, and 400×, Oxford-IIIT Pet, and Chest X-ray, respectively. The only exception occurs on BreakHis at 200×, where the w/o Att variant obtains a slightly higher correlation than the complete model (0.872 versus 0.859). Nevertheless, the complete model remains superior in AUC, ACC, and ECE. Overall, the ablation results demonstrate that bidirectional feature flow contributes to the integration of complementary cross-scale information, while noise-aware attention suppresses unreliable features before fusion. Their combination produces the most accurate, calibrated, and generally uncertainty-informative predictions.

Importantly, adding bidirectional feature flow provides larger performance gains than adding noise-aware attention alone. The w/o Att variant outperforms the w/o Bidir variant in ACC on all six settings and in AUC on five of six (the exception is Oxford-IIIT Pet). For example, the AUC/ACC differences are 0.049/0.061, 0.058/0.064, and 0.052/0.066 on BreakHis at 40×, 100×, and 200×, respectively. On Oxford-IIIT Pet and Chest X-ray, the corresponding differences are −0.004/0.028 and 0.023/0.033. These results indicate that

bidirectional cross-scale information exchange makes the larger contribution to discriminative performance, while noise-aware attention provides an additional complementary improvement by suppressing unreliable features.

4.6.2. Uncertainty Threshold

The uncertainty threshold controls the tolerance for uncertainty in the final decision. A lower threshold imposes a stricter reliability criterion, whereas a higher threshold allows predictions with greater uncertainty to contribute to the evaluation. Figure 6 shows the ACC and AUC of the proposed method when the threshold varies from 0 to 1 on the six evaluated datasets.

Both metrics generally decrease as the uncertainty threshold increases. This trend is consistent across the four BreakHis subsets, Oxford-IIIT Pet, and Chest X-ray. The decline is more apparent at larger threshold values, particularly from 0.5 to 1.0. These results indicate that highly uncertain predictions are more likely to be incorrect and can reduce classification performance when they are accepted. The proposed method remains relatively stable at lower thresholds, which suggests that its subjective-logic evidence fusion produces reliable uncertainty estimates.

A threshold of 0.5 is selected for all the experiments. At this value, the method preserves high ACC and AUC across all the datasets while avoiding overly permissive acceptance of uncertain predictions. Lower thresholds provide only limited performance gains in several datasets but may reject a larger number of predictions. In contrast, thresholds above 0.5 consistently reduce both metrics. Therefore, 0.5 provides a balanced operating point between classification performance and uncertainty tolerance.

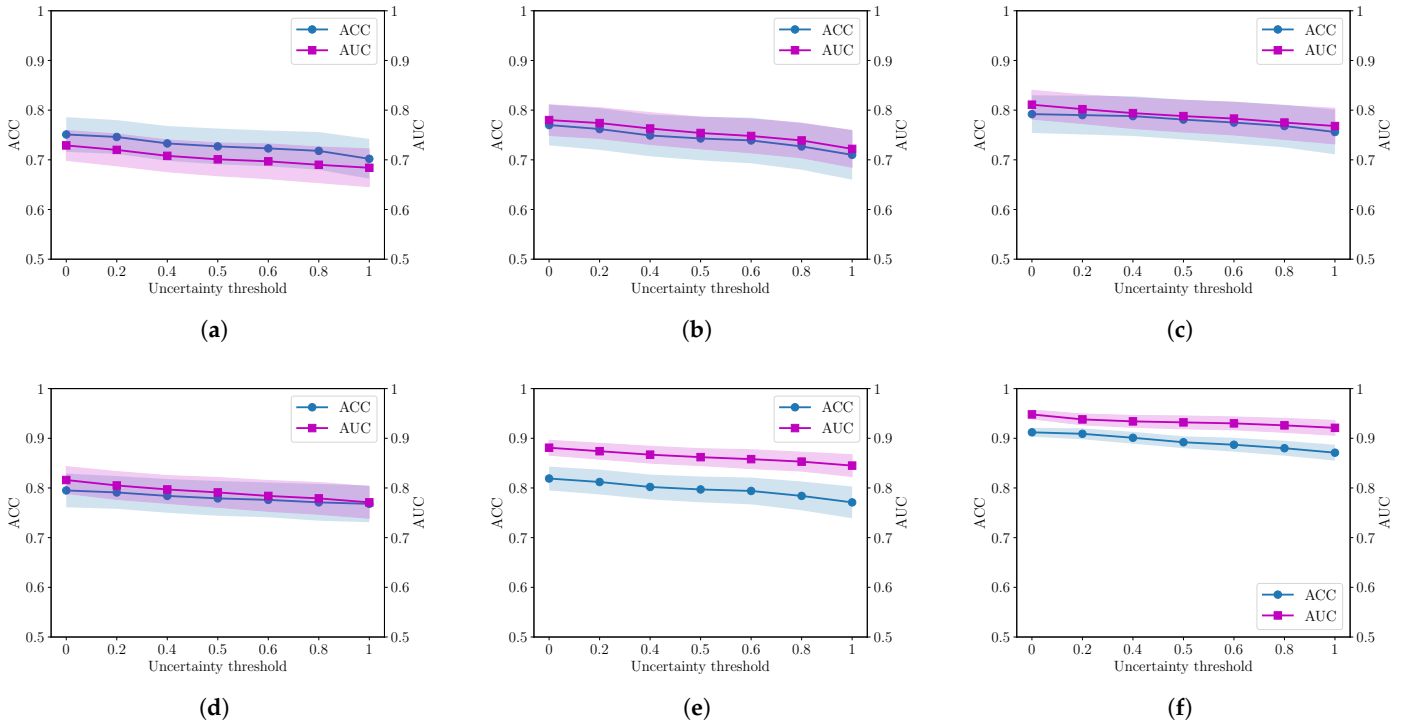


Figure 6. Classification performance on different real-world datasets with varying uncertainty thresholds. (**a**) BreakHis 40×; (**b**) BreakHis 100×; (**c**) BreakHis 200×; (**d**) BreakHis 400×; (**e**) Oxford-IIIT Pet; (**f**) Chest X-ray.

4.6.3. Robustness to Gaussian Noise

The robustness to Gaussian noise is evaluated by adding zero-mean Gaussian noise with a standard deviation ranging from 0 to 10^9 to the input images. All the methods use two pseudo-views. For SE-PML and the proposed method, noise is injected separately into either the top or bottom branch to examine the effect of branch-wise noise propagation. Figure 7 reports the classification accuracy under different noise levels on the six evaluation datasets.

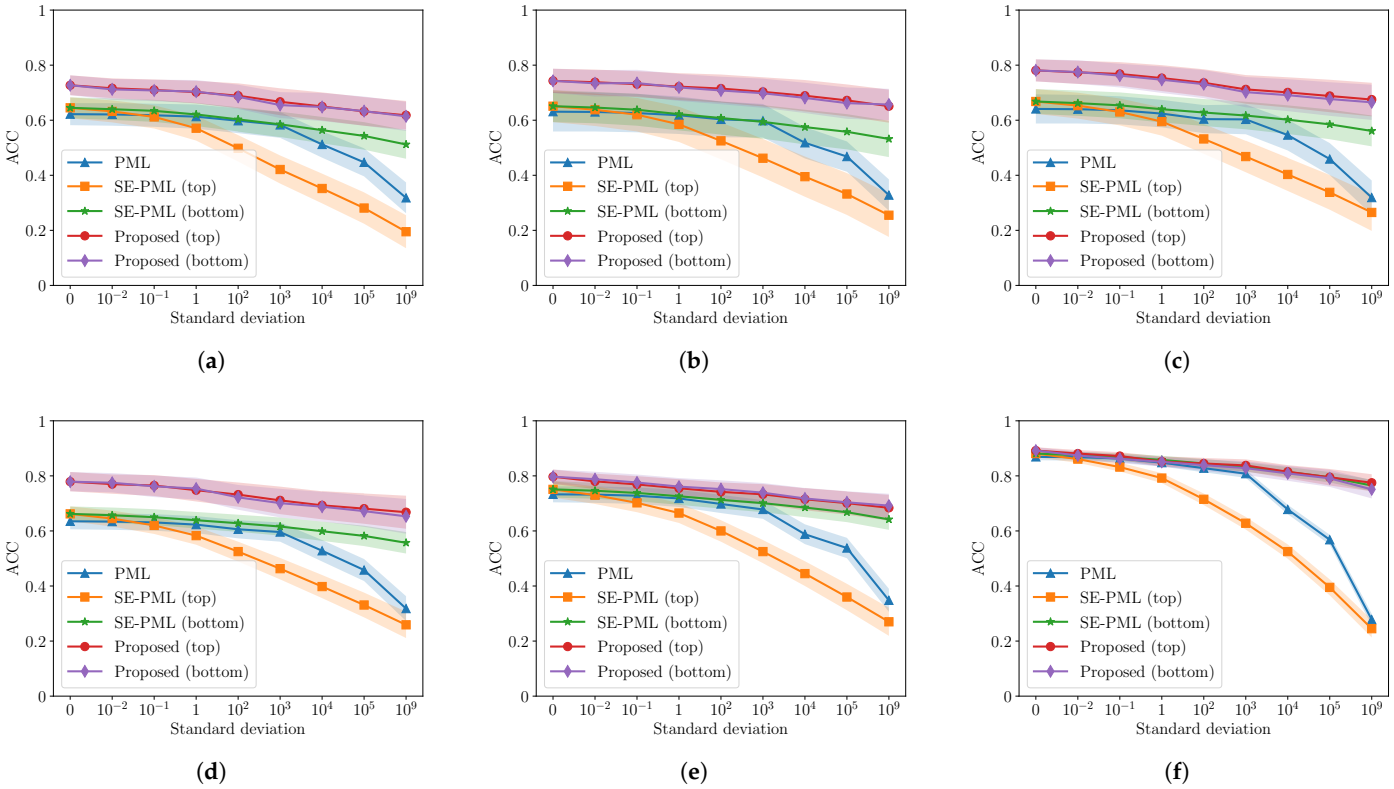


Figure 7. Classification accuracy on different real-world datasets with varying noise levels. (**a**) BreakHis 40×; (**b**) BreakHis 100×; (**c**) BreakHis 200×; (**d**) BreakHis 400×; (**e**) Oxford-IIIT Pet; (**f**) Chest X-ray.

PML shows a consistent reduction in accuracy as the noise level increases. This result is expected because PML relies on pseudo-views generated by separate feature streams and lacks an explicit mechanism for suppressing corrupted features. The degradation becomes more pronounced at high noise levels.

SE-PML is highly sensitive to noise injected into the top branch. Its accuracy declines sharply as the noise level increases, and the degradation is substantially larger than that observed when noise is injected into the bottom branch. For example, on BreakHis 40×, SE-PML with top-branch noise decreases from 0.645 under clean input to 0.195 at the highest noise level. In contrast, its bottom-branch-noise case retains an accuracy of 0.5. Similar trends are observed across the remaining datasets. This difference supports the identified limitation of SE-PML: its top-down feature flow propagates corrupted high-resolution information to lower-resolution branches, which amplifies the effect of top-branch noise.

The proposed method maintains higher accuracy than PML and SE-PML under both top- and bottom-branch noise. Its curves show only a gradual performance decrease as the noise level increases. More importantly, the gap between the top- and bottom-branch noise cases remains small. This behavior indicates that the proposed bidirectional feature flow

does not allow noise from a single branch to dominate the fused representation. Instead, features from the unaffected branch provide complementary information during fusion.

The result is particularly clear on Chest X-ray. While PML and SE-PML with top-branch noise decrease substantially at high noise levels, the proposed method maintains an accuracy close to 0.75 when noise is injected into either branch. Similar robustness is observed on the BreakHis subsets, where the proposed method remains above approximately 0.60 at the highest noise level. These results support the proposed noise-aware spatial-channel attention mechanism. By estimating feature reliability during channel-wise and spatial fusion, it suppresses corrupted responses before they propagate across branches. Consequently, the proposed method mitigates the noise-propagation limitation of SE-PML and provides more stable predictions under severe input corruption.

5. Conclusions

This paper presented an improved pseudo-multiview learning framework via dual-sampler bidirectional feature flow for effective and noise-resilient image classification. The method addresses a key limitation of the existing pseudo-multiview learning frameworks: noise propagation in unidirectional multiscale feature flow.

The proposed framework generates pseudo-views at multiple resolutions and introduces a dual-sampler module for bidirectional feature exchange. This design allows high-resolution branches to provide detailed spatial information and enables rapidly convergent low-resolution branches to provide stable contextual guidance to higher-resolution branches. In addition, the proposed noise-aware spatial-channel attention module evaluates both feature importance and reliability during cross-scale fusion. It suppresses unreliable spatial regions and feature channels before corrupted information can affect other branches. The resulting pseudo-view evidence is fused using subjective logic to produce uncertainty-aware predictions.

Experiments on BreakHis at four magnification levels, Oxford-IIIT Pet, and Chest X-ray demonstrated that the proposed method consistently outperformed PML, SE-PML, and single-view uncertainty-aware baselines in AUC and classification accuracy. The convergence analysis further showed faster and more stable optimization than the baseline methods. Moreover, robustness experiments with branch-wise Gaussian noise showed that the proposed method limits cross-scale noise propagation and maintains stronger performance under severe input corruption.

However, the convergence benefit attributed to low-resolution branches is supported by the overall training curves and ablation results rather than a branch-wise analysis. Future work will examine branch-wise loss and accuracy curves and feature representations across training stages to verify this mechanism directly. Moreover, the noise experiments are limited to synthetic Gaussian noise injected into one branch at a time. Evaluating other noise types, noise in multiple branches, and real corrupted images, as well as designing a framework that is robust across them, is an important direction for future work. Adaptive pseudo-view selection will also be explored to further reduce unnecessary computation and extend the proposed framework to dense prediction tasks, such as object detection and semantic segmentation.

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Data Availability Statement: Data available in publicly accessible repositories. BreakHis: https://ieee-dataport.org/documents/breakhis-dataset (accessed on 12 April 2026). Oxford-IIIT Pet: https://www.robots.ox.ac.uk/~vgg/data/pets/ (accessed on 12 April 2026). Chest X-ray: https://data.mendeley.com/datasets/rscbjbr9sj/2 (accessed on 12 April 2026).

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